JOURNAL OF AGRICULTURAL AND FOOD CHEMISTRY

Article

pubs.acs.org/JAFC

Characterization of the Typical Potent Odorants in Chinese Roasted Sesame-like Flavor Type Liquor by Headspace Solid Phase Microextraction−Aroma Extract Dilution Analysis, with Special Emphasis on Sulfur-Containing Odorants

Sha Sha,[†] Shuang Chen,[†] Michael Qian,[§] Chengcheng Wang,[†] and Yan Xu*[,†]

[†]State Key Laboratory of Food Science & Technology, Key Laboratory of Industrial Biotechnology of Ministry of Education & School of Biotechnology, Jiangnan University, Wuxi, Jiangsu, China 214122

[§]Department of Food Science & Technology, Oregon State University, Corvallis, Oregon 97331, United States

ABSTRACT: The aroma profile of Chinese roasted sesame-like flavor type liquor was investigated by means of headspace solid phase microextraction−aroma extract dilution analysis (HS-SPME-AEDA). Sixty-three odor-active regions were detected by HS-SPME-AEDA with flavor dilution (FD) factors >5, and 58 of these were further identified. Among them, ethyl hexanoate, 2-furfurylthiol, dimethyl trisulfide, 3-methylbutanal, ethyl butanoate, ethyl 2-methylbutanoate, ethyl pentanoate, and ethyl 4-methylpentanoate appeared with the highest FD factors. In particular, eight sulfur-containing odorants were identified to be potentially important to roasted sesame-like flavor type liquor. The concentration of these odor-active compounds was further quantitated by combination of four different quantitative measurements, and 36 odorants had concentrations higher than their corresponding odor thresholds. On the basis of the odor activity values (OAVs), 2-furfurylthiol (OAV 1182), dimethyl trisulfide (OAV 220), β-damascenone (OAV 116), and methional (OAV 99) could be responsible for the unique aroma of roasted sesame-like flavor type liquor. An aroma recombination model prepared by mixing 36 aroma compounds with OAVs >1 showed a good similarity to the aroma of the original roasted sesame-like flavor type liquor. For the first time, 2-furfurylthiol was determined to be a typical potent odorant in roasted sesame-like flavor type liquor by omission study.

KEYWORDS: *Chinese roasted sesame-like flavor type liquor, HS-SPME-AEDA, OAV, aroma recombinate, 2-furfurylthiol*

INTRODUCTION

Chinese liquor (*baijiu*) is a traditional indigenous distilled spirit in China, normally with an alcohol content of 40−55% (by vol). It is one of the most popular alcoholic beverages in China. Chinese liquor is typically made from sorghum or a mixture of wheat, barley, corn, rice, and sorghum with a natural mixed culture starter, Daqu, as fermentation starter.[1,2] The aroma profile of Chinese liquor is greatly influenced by the Daqu qualities, liquor-making processes, and fermentations.[3,4] On the basis of aroma characteristics, Chinese liquors are generally classified into 11 flavor types, including strong, light, soy sauce, sweet and honey, chixiang, complex, herblike, fengxiang, laobaigan, texiang, and roasted sesame-like flavor type liquors.[5] Among them, roasted sesame-like flavor type liquor has gained popularity due to its unique roasted sesame-like aroma.

Aroma is one of the most important sensory components that contribute to liquor quality and consumer acceptance. Data on the volatile composition of roasted sesame-like flavor type liquor were published as early as 1986, and over 163 volatile components were identified.[6] Subsequently, many studies were undertaken to analyze the entire volatile compounds of roasted sesame-like flavor type liquor, and >250 volatiles were identified.[7,8] Among these volatiles, sulfur-containing compounds gained special attention. Up to now, >20 sulfur compounds have been identified in roasted sesame-like flavor liquors.[9,10] However, due to the lack of quantitative and odor activity value (OAV) data, the contribution of these sulfur compounds to the overall aroma was not verified. Key odorants for the typical roasted sesame-like odor are still not clear.

A meta-analysis of the chemical odor codes of >220 food samples by Dunkel et al.[11] shows that only a small subset of volatiles (key odorants) in food constitute the chemical odorant space. Therefore, identification of key odorants from the complex mixture of volatile components is the most important task in flavor analysis.[12] Gas chromatography−olfactometry (GC-O) with aroma extract dilution analysis (AEDA) was a useful tool for screening of potential important odorants in food.[13,14] Key odorants can be further identified by OAV, aroma recombination, and omission test.[15,16] Very recently, Zheng et al.[8] investigated the key aroma compounds in roasted sesame-like flavor type liquor on the basis of AEDA, OAV, aroma recombination, and omission test, and 26 key odorants were identified in roasted sesame-like flavor type liquor. However, most of the key aroma compounds identified in the study were similar to those identified previously in other types of Chinese liquor.[3,5,17,18] Because none of compounds with roasted sesame-like odor has been identified in roasted sesame-like flavor type liquor up to now, the characteristic odorants responsible for the roasted sesame-like note are still missing.

Received: September 23, 2016
Revised: December 12, 2016
Accepted: December 17, 2016
Published: December 17, 2016

ACS Publications © 2016 American Chemical Society

DOI: 10.1021/acs.jafc.6b04242
J. Agric. Food Chem. 2017, 65, 123−131

Downloaded from pubs.acs.org/jafcau/article-pdf/65/1/123/1411259/jf6b04242.pdf by JIANGNAN UNIV user on 18 October 2026

Therefore, the aims of the present study were (i) to identify the potential important odorants in the roasted sesame-like flavor type liquor by HS-SPME-AEDA, (ii) to quantify the aroma compounds in roasted sesame-like flavor type liquor sample by multiple quantitation methods, (iii) to determine the importance of each aroma compounds on the basis of OAVs, and (iv) to verify the results by means of aroma reconstitution and omission experiment.

MATERIALS AND METHODS

Chemicals. All chemical standards of the aroma compounds used in this study were of GC grade, with at least 95% purity. Dimethyl sulfide, *S*-methyl thioacetate, dimethyl trisulfide, 2-furfurylthiol, methional, ethyl 2-mercaptoacetate, ethyl 3-(methylthio)propanoate, methionol, ethyl hexanoate, ethyl octanoate, ethyl butanoate, 3-methylbutanal, ethyl 3-methylbutanoate, ethyl 2-methylbutanoate, 3-methylbutyl acetate, 2-methylpropanoic acid, ethyl acetate, 3-methylbutanoic acid, pentanoic acid, ethyl 2-phenylacetate, ethyl lactate, 2-methylpropanol, 1-propanol, heptanoic acid, butanoic acid, 3-methylbutyl butanoate, 1-hexanol, hexanoic acid, furfural, 2-phenylethyl acetate, 3-methylbutyl hexanoate, 4-methylphenol, ethyl decanoate, 2-phenylethyl hexanoate, ethyl dodecanoate, 2-phenylethyl butanoate, ethyl nonanoate, 3-methylbutanol, ethyl heptanoate, 2-phenylethanol, 2-methylpropyl hexanoate, ethyl 2-furoate, 2-nonanone, ethyl benzoate, phenol, 4-ethylphenol, 3-methylbutyl octanoate, furfuryl hexanoate, 2-undecanone, nonanal, ethyl 2-methylpropanoate, propyl hexanoate, benzaldehyde, naphthalene, and 3-methylbutyl acetate were purchased from Sigma-Aldrich Co., Ltd. (Shanghai, China). Acetic acid, butanoic acid, and ethyl propanoate were supplied by Alfa Aesar (Tianjin, China). 2-Octanol (internal standard, IS1), 2,2-dimethylpropanoic acid (IS2), methyl hexanoate (IS3), octyl propanoate (IS4), isopropyl disulfide (IS5), and 4-(methylthio)-1-butanol (IS6) were used as internal standards purchased from Sigma-Aldrich Co., Ltd. A C_5−C_{30} *n*-alkane mixture (Sigma-Aldrich,) was employed for determination of linear retention indices (RIs). Sodium chloride (NaCl) and anhydrous sodium sulfate (Na_2SO_4) were purchased from China National Pharmaceutical Group Corp. (Shanghai, China). Diethyl ether from ANPEL Scientific Instrument Co., Ltd. (Shanghai, China) was freshly distilled before use.

Sample. A commercial roasted sesame-like flavor type liquor named "Shangpin, SP" (53% ethanol by volume) was used in this study. It was manufactured by Jingzhi Liquor Co. Ltd., (Shandong, China) according to the national standard of roasted sesame-like flavor type liquor (GB/T 2082-2007). The representative of this sample was confirmed by sensory evaluation with a sensory panel composed of five national Chinese liquor judges. The sample was stored at 4 °C until analysis.

Gas Chromatographic−Olfactometric and −Mass Spectrometric Analysis. *Volatile Extraction with HS-SPME.* The sample preparation and HS-SPME technique were performed according to the methods described previously.[18] Liquor (50 mL) was diluted to 10% ethanol by volume with boiled ultrapure water. Ten milliliters of diluted liquor and a 4 mm Teflon-coated stir bar were added into a 20 mL glass via, which was flushed with argon before sealing with a septum screw top. Samples were equilibrated at 45 °C in a water bath for 15 min prior to analysis. After equilibration, a 2 cm fiber coated with 50/30 μm divinylbenzene/carboxen/polydimethylsiloxane (DVB/CAR/PDMS; Supelco, Bellefonte, PA, USA) was exposed to the headspace of the vial for 30 min at the same temperature. The fiber was then introduced into the GC injection port at 250 °C for 5 min of desorption.

Direct GC-O. According to the method of Du et al.,[19] a 1 m deactivated silica column (0.25 mm i.d.) was installed on an Agilent 6890 gas chromatograph equipped with an Agilent 5975 mass-selective detector (MSD) and an olfactometer (ODP 2, Gerstel, Mülheim, Ruhr, Germany). The flow rate of the helium carrier gas was 2 mL/min. The injection port temperature was 250 °C, and the oven temperature was 200 °C (isothermal). The configuration allows for the evaluation of global odor of the extracts without chromatographic separation, and the entire analysis was completed in <30 s. Two trained panelists smelled the odor of the unseparated HS-SPME extract at the GC sniffing port. Then, panelists opened a 40 mL vial to smell the original liquor. Panelists rated the similarity between the unseparated HS-SPME extract and the original liquor odors using a 6-point scale ranging from 0 (no similarity) to 5 (exactly the same).

GC-O and GC-MS Analysis. GC-O and GC-MS analyses were performed on an Agilent 6890 gas chromatograph equipped with an Agilent 5975 mass-selective detector (MSD) and an olfactometer (ODP 2, Gerstel). Samples were analyzed on a DB-FFAP column (60 m × 0.25 mm i.d., 0.25 μm film thickness, J&W Scientific) and a DB-5 column (30 m × 0.25 mm i.d., 0.25 μm film thickness, J&W Scientific). The GC-MS conditions in this study were previously reported.[20] The column carrier gas was helium at a constant flow rate of 2 mL/min. The sniffing port had a split ratio of 1:1. The injector temperature was set at 250 °C. The oven temperature programs were as follow: 45 °C for 2 min, raised at 6 °C/min to 230 °C, and held at 230 °C for 10 min (DB-FFAP); and 45 °C for 2 min, raised at 6 °C/min to 270 °C, and held at 270 °C for 10 min (DB-5). The temperature of the olfactory port was kept at 250 °C. Mass spectra in the electron ionization mode (EI) were recorded at 70 eV ionization energy. The temperature of the ion source was 230 °C, and the mass range was from 30 to 350 amu.

Olfactometry analysis was carried out by four trained panelists (two females and two males), three graduate students and one teacher from the Laboratory of Brewing Microbiology and Applied Enzymology at Jiangnan University. The panelists were trained for 2 months in GC-O using at least 30 odor-active reference compounds in a concentration 10 times above their odor thresholds in air. During a GC run, a panelist placed his/her nose close to and above the top of the sniffing port and recorded the odor of the chromatographic effluent as well as retention time. Analyses were repeated in duplicate by each panelist.[21]

AEDA. When working with the HS-SPME, there is no liquid extract because the analytes are retained on the fiber. Therefore, the usual AEDA cannot be applied. According to Feng et al.,[22] the aroma concentrate of the SP, extracted by HS-SPME, was stepwise diluted by different spilt ratios, varying from 5:1, 10:1, 25:1, 50:1, 100:1, 200:1, 400:1 to 600:1. AEDA was performed on the DB-FFAP column as above. The FD factor was defined as the dilution step at which a compound was detected at least five of the eight times. By definition, the FD factor obtained for each odorant in the HS-SPME-AEDA is equal to the highest spilt ratio.

Identification of Aroma Compounds. The identification of the odorants was carried out by comparison of their odors, mass spectra, and linear retention indices (LRI) with those of pure reference compounds. LRIs of the odorants were calculated from the linear retention times of *n*-alkanes (C_5−C_{30}) in both DB-FFAP and DB-5 columns, according to a modified Kovats method.[23]

Quantitative Analysis of Aroma Compounds. *Gas Chromatography with Flame Ionization Detector (GC-FID).* Quantitative analysis of major compounds (ethyl acetate, ethyl butanoate, ethyl hexanoate, ethyl heptanoate, ethyl octanoate, ethyl lactate, 1-propanol, 2-methylpropanol, 3-methylbutanol, 1-butanol, and 1-hexanol) was carried out by GC-FID.[24] A liquor sample was spiked with 10 μL of internal standard solution (IS1) to a final concentration 176 mg/L. One microliter of disposed sample was directly injected into the GC in split mode (split ratio = 37:1). Nitrogen was used as carrier gas at a constant flow rate of 1 mL/min. Separation was performed on a DB-Wax column (30 m × 0.25 mm i.d. × 0.25 μm film thickness; J&W Scientific). The oven temperature was initially set at 60 °C for 3 min, ramped at 5 °C/min to 150 °C for 5 min, and then increased to 230 °C at 10 °C/min for 5 min. The injector and detector temperatures were set at 250 °C. Individual standard stock solution was mixed and then diluted with ethanol aqueous solution (53%, vol) to a serial concentration to set up the calibration curve. The standard calibration curve working solution was analyzed by GC-FID, as was performed for the liquor sample. The calibration curves were obtained from Chemstation software (Agilent Technologies Inc.) and used for the

Downloaded from pubs.acs.org/jafcau/article-pdf/65/1/123/1411259 6/jf6b04242.pdf by JIANGNAN UNIV user on 18 October 2026

DOI: 10.1021/acs.jafc.6b04242
J. Agric. Food Chem. 2017, 65, 123−131

Table 1. Odor-Active Volatiles Identified in SP Roasted Sesame-like Flavor Type Liquor by HS-SPME-AEDA

no.	LRI FFAP[a]	LRI DB-5[b]	compound identified	odor quality[c]	FD factor[d]	identification[e]
1	908	605	ethyl acetate	pineapple	100	RI, MS, aroma
2	922	516	dimethyl sulfide	cooked onion	5	RI, MS, aroma
3	928	652	3-methylbutanal	green	400	RI, MS, aroma
4	935	708	ethyl propanoate	fruity	100	RI, MS, aroma
5	964	753	ethyl 2-methylpropanoate	fruity, sweet	100	RI, MS, aroma
6	1018	702	*S*-methyl thioacetate	rotten cabbage	100	RI, MS, aroma
7	1027	810	ethyl butanoate	fruity	400	RI, MS, aroma
8	1044	536	1-propanol	fruity	100	RI, MS, aroma
9	1049	850	ethyl 2-methylbutanoate	fruity	400	RI, MS, aroma
10	1061	858	ethyl 3-methylbutanoate	apple	100	RI, MS, aroma
11	1090	620	2-methylpropanol	malty	25	RI, MS, aroma
12	1132	909	ethyl pentanoate	fruity	400	RI, MS, aroma
13	1179	915	methyl hexanoate	floral	25	RI, MS, aroma
14	1190	965	ethyl 4-methylpentanoate	fruity	400	RI, MS, aroma
15	1211	768	3-methylbutanol	malty	25	RI, MS, aroma
16	1244	1016	ethyl hexanoate	floral	600	RI, MS, aroma
17	1300	968	1-octen-3-one	mushroom	5	RI, MS, aroma
18	1318	1101	propyl hexanoate	fruity	200	RI, MS, aroma
19	1336	925	2,6-dimethylpyrazine	nutty	100	RI, MS, aroma
20	1338	821	ethyl lactate	fruity	5	RI, MS, aroma
21	1350	865	1-hexanol	floral	25	RI, MS, aroma
22	1385	962	dimethyl trisulfide	sulfur, rotten cabbage	400	RI, MS, aroma
23	1403	1000	1,3,5-trimethylpyrazine	baked	5	RI, MS, aroma
24	1408	1202	ethyl octanoate	fruity	25	RI, MS, aroma
25	1412	1080	2,6-diethylpyrazine	baked	5	RI, MS, aroma
26	1419	994	2,3,5-trimethylpyrazine	baked	25	RI, MS, aroma
27	1428	1242	3-methylbutyl hexanoate	fruity	100	RI, MS, aroma
28	1434	1079	2,5-dimethyl-3-ethylpyrazine	baked	100	RI, MS, aroma
29	1438	1080	2-furfurylthiol	roasted sesame seeds	400	RI, MS, aroma
30	1451	834	furfural	butter	5	RI, MS, aroma
31	1454	625	acetic acid	vinegar	5	RI, MS, aroma
32	1461	908	methional	cooked potato	25	RI, MS, aroma
33	1471	915	2-acetylfuran	sweet	25	RI, MS, aroma
34	1489	nd	unknown	stink	5	unknown
35	1502	959	benzaldehyde	fruity	5	RI, MS, aroma
36	1513	nd	unknown	musty	10	unknown
37	1528	1068	ethyl 2-mercaptoacetate	cooked vegetable	100	RI, MS, aroma
38	1540	nd	unknown	musty	100	unknown
39	1562	963	5-methylfurfural	baked	25	RI, MS, aroma
40	1590	1098	ethyl 3-(methylthio)propanoate	sulfur, rotten cabbage	5	RI, MS, aroma
41	1594	1389	hexyl hexanoate	apple, peach	10	RI, MS, aroma
42	1614	1041	2-acetyl-5-methylfuran	baked	5	RI, MS, aroma
43	1620	812	butanoic acid	rancid	10	RI, MS, aroma
44	1646	1168	ethyl benzoate	fruity	10	RI, MS, aroma
45	1675	873	3-methylbutanoic acid	rancid	100	RI, MS, aroma
46	1678	1173	diethyl butanedioate	sweet	25	RI, MS, aroma
47	1688	1189	terpineol	floral	200	RI, aroma
48	1724	nd	unknown	mint, solvent	10	unknown
49	1736	1184	naphthalene	musty	5	RI, MS, aroma
50	1745	980	methionol	cooked vegetable	100	RI, MS, aroma
51	1795	1243	ethyl 2-phenylacetate	rosy, floral	200	RI, MS, aroma
52	1824	1246	2-phenylethyl acetate	floral	200	RI, MS, aroma
53	1830	1390	*β*-damascenone	sweet, candy	100	RI, MS, aroma
54	1843	1087	guaiacol	spicy, clove	5	RI, MS, aroma
55	1851	990	hexanoic acid	rancid	50	RI, MS, aroma
56	1862	1439	geosmin	earthy	100	RI, MS, aroma
57	1920	1118	2-phenylethanol	floral	200	RI, MS, aroma
58	1969	990	heptanoic acid	rancid	5	RI, MS, aroma
59	1986	984	phenol	medicinal	5	RI, MS, aroma

Downloaded from pubs.acs.org/jafcau/article-pdf/65/1/123/1411259/jf6b04242.pdf by JIANGNAN UNIV user on 18 October 2026

DOI: 10.1021/acs.jafc.6b04242
J. Agric. Food Chem. 2017, 65, 123−131

Table 1. continued

no.	LRI FFAP[a]	LRI DB-5[b]	compound identified	odor quality[c]	FD factor[d]	identification[e]
60	2038	1358	γ-nonanolactone	coconut	5	RI, MS, aroma
61	2101	1074	*m*-cresol	leather	5	RI, MS, aroma
62	2123	nd	unknown	medicinal	5	unknown
63	2202	1168	4-ethylphenol	smoky	5	RI, MS, aroma

[a]FFAP, linear retention index on DB-FFAP. [b]DB-5, linear retention index on DB-5. [c]Odor quality perceived at the sniffing port. [d]FD factor, flavor dilution factor, defined as the highest spilt ratio at which the odorant could be perceived by HS-SPME-AEDA. [e]Identification based on LRI (linear retention index) or MS (mass spectrometry) or odor description by comparison to the pure standard.

calculation of volatiles in the sample. The sample was performed in triplicate.

Liquid–Liquid Microextraction–GC-MS (LLME-GC-MS). Fatty acids were quantified by LLME-GC-MS according to the method of Wang et al.[17] A diluted liquor sample (18 mL) with 6 μL of internal standard solution (IS2, 3.41 mg/L final concentration in ethanol) was saturated with NaCl and then extracted for 3 min with 1 mL of redistilled diethyl ether. The GC-MS conditions were set as for GC-O analysis on a DB-FFAP column (60 m × 0.25 mm i.d., 0.25 μm film thickness, J&W Scientific). A calibration curve working solution was prepared in aqueous ethanolic solution (53%, vol) as described above. Six microliters of IS2 solution was added to each working solution and then analyzed by LLME-GC-MS. The calibration curves were obtained from Chemstation software (Agilent Technologies Inc.) and used for calculation of volatiles in the sample. The SP liquor was performed in triplicate.

Headspace Solid-Phase Microextraction–Gas Chromatography–Mass Spectrometry. Most minor compounds were quantitated using the method proposed and validated by Gao et al.[20] Liquor sample was diluted with Milli-Q water (Millipore, Bedford, MA, USA) to a final concentration of 10% ethanol by volume. A total of 8 mL of diluted solution with 10 μL of internal standard solution (IS3 and IS4, with concentrations of 87.8 and 163 mg/L in ethanol, respectively) was put into a 20 mL screw-capped vial and then saturated with NaCl. An automatic headspace sampling system (MultiPurposeSample MPS 2 with a SPME adapter, from Gerstel) with a 2 cm, 50/30 μm, DVB/CAR/PDMS fiber (Supelco Inc.) was used to extract volatile odorants. The GC-MS conditions were set as for GC-O analysis on a DB-FFAP column (60 m × 0.25 mm i.d., 0.25 μm film thickness, J&W Scientific) described previously.[20] Standard calibration curves were developed for each individual volatile and were used to calculate the concentrations of volatiles in SP. Triplicate analysis was performed for the sample.

HS-SPME-GC-Pulsed Flame Photometric Detector (HS-SPMEGC-PFPD). According to the method published previously,[25−27] sulfur volatile analyses were performed using an Agilent 7890A gas chromatograph equipped with a pulsed flame photometric detector (OI Analytical model 5380, OI Analytical Co., College Station, TX, USA). Separation was achieved using a DB-FFAP column (30 m × 0.32 mm i.d., 1 μm film thickness, J&W Scientific). The helium column flow was 2 mL/min. The oven temperature was programmed at 35 °C for a 3 min initial hold, ramped to 150 °C at 10 °C/min, held 5 min, increased at 20 °C/min to a final temperature of 220 °C, and held 3 min. The GC injection temperature was 250 °C, and the detector temperature was 250 °C. Sulfur gate time was 6−24.9 ms, and pulse frequency was approximately 3 pulses/s. Standard calibration curves for sulfur volatiles were obtained by adding authentic sulfur standards of known concentration to aqueous ethanolic solution (53%, vol). A series of concentration levels of the standards in aqueous ethanolic solution was prepared. Coupled with the internal standards, the added sulfur compounds were extracted with HS-SPME, as performed for the sample. Standard calibration curves were developed for each individual sulfur volatile and were used to calculate the concentrations of sulfur volatiles in sample. Triplicate analyses were performed for the sample. The limits of detection (LOD) were calculated as the analyte concentration of a standard that produced a signal-to-noise ratio of 3.

Determination of Odor Thresholds. On the basis of the method previously described,[28] the orthonasal thresholds of the odorants were determined by a forced-choice test at seven concentration steps. A certain amount of the odorant in ethanol was pipetted into a Teflon vessel containing 50 mL of hydroalcoholic solution at 46% ethanol by volume, stirred for 2 min, and stepwise diluted (1:3 by volume, with the hydroalcoholic solution). Triangular series including one glass of the dilution and two glasses of hydroalcoholic solution were prepared. All of the series were labeled with random four-digit numbers and presented in decreasing concentrations. A sensory panel consisting of 32 panelists was asked to sniff each triangular series and select the differing one. The minimum concentration that the assessors correctly selected and the maximum concentration incorrectly selected were recorded. The odor threshold of each odorant was calculated by using the formulas

$$OT_i = \sqrt{C_X \times C_{X+1}}$$

where OT_i is the individual recognition/detection odor threshold of each assessor, C_X is the lowest concentration of the odorant that was correctly selected by the assessor in a series, and C_{X-1} is the highest concentration of the odorant that was incorrectly selected by the assessor, and

$$OT_P = \sqrt[n]{\prod_{n}^{X-1} OT_i}$$

where OT_P is the recognition/detection odor threshold of the panel, n is the number of assessors, and $\Pi_{n-1}^{n}\ OT_i$ is the product of individual recognition/detection odor thresholds.

Sensory tests were performed at 21 ± 1 °C in a sensory room, and all of the panelists were national liquor tasters and had prior sensory training in Chinese liquor evaluation.

Descriptive Profile Tests. Sensory analyses were performed in a panel room at 21 ± 1 °C;[29] the descriptive aroma profiles were performed as described with two specific training sessions. In the first one, different aroma standards and SP liquor sample were presented and discussed. From this session, eight aroma terms were selected for further descriptive analysis. In session two, panelists scored the intensity of each attribute on one six-point scale from 0 to 5. Eight aroma terms were defined as follows: ethanol for alcoholic note, β-damascenone for sweet note, 1,3,5-trimethylpyrazine for baked note, ethyl hexanoate for fruity note, acetic acid for acid note, dimethyl trisulfide for rotten vegetable note, and roasted sesame seed for roasted sesame. After the training, the overall aroma profile of roasted sesame-like flavor type liquor was evaluated by the panel.

Aroma Recombination of SP. A hydroalcoholic solution at 53% ethanol by volume was used as the matrix for recombination. All 36 odorants with OAVs >1 simultaneously were dissolved in the matrix in their natural concentrations and then equilibrated for 10 min at ambient temperature; thus, a complete recombinate was obtained. This recombinate of 20 mL was finally presented in a glass covered with aluminum foil. Besides, a glass of SP liquor was also prepared for the following sensory evaluation. Descriptive profile tests were performed by 10 assessors (5 males and 5 females, 32 years old on average), 4 of whom were from the GC-O analysis and the other 6 were laboratory staff members familiar with the sensory attributes of

Downloaded from pubs.acs.org/jafcau/article-pdf/65/1/123/1411259 6/jf6b04242.pdf by JIANGNAN UNIV user on 18 October 2026

DOI: 10.1021/acs.jafc.6b04242
J. Agric. Food Chem. 2017, 65, 123−131

Chinese liquors. Eight attributes were chosen as the most relevant odor to describe the overall aroma: roasted sesame, sweet, baked, fruity, acid, floral, alcoholic, and rotten vegetable. The assessors were asked to evaluate the odor intensity of these attributes on a 6-point scale from 0 to 5. The results obtained from 10 assessors were averaged and finally plotted in a spider web diagram. All of the tests were conducted in a sensory panel room at 20 ± 1 °C.

Omission Experiments. A triangle test was performed to determine the significance of one odorant. The testing samples were arranged in a random four-digit code, and the test was repeated in triplicate. The omission experiment was performed by 10 assessors. All of the assessors were previously trained in orthonasal odor perception and participated regularly in sensory evaluation. The assessors were asked to sniff the samples and estimate the differing one. The significance of the difference was evaluated according to the method previously described.[30] The sensory data were analyzed by one-way analysis of variance (ANOVA) by use of SPSS15.0 (SPSS Inc., Chicago, IL, USA).

RESULTS AND DISCUSSION

Odor-Active Compounds Determined by HS-SPME-GC-O. First, the volatiles were extracted by HS-SPME, which is a quick, solvent-free, and quite simple technique that can avoid the loss of highly volatile components.[31,32] The global aromas of the extracts were evaluated using the unseparated GC-O method. The global aroma profile of HS-SPME extracts was similar to that of the original liquor, which has accounted for 4.0 in a 6-point (0−5) scoring method. The typical roasted sesame-like odor has been clearly detected by direct GC-O analysis. The results suggested that collected HS-SPME volatiles generated an odor that was representative of the original sample. Then the HS-SPME extracts were submitted to AEDA. A total of 63 aroma active regions were detected with FD factors ranging from 5 to 600 (Table 1; Figure 1). It can be seen that fruity and floral aroma characters are the major contributors to SP liquor. Among them, the highest FD factor was found for a fruity compound (**16**; FD of 600) eluting with a retention index 1244, followed by a green-smelling odorant (**3**; FD of 400). In addition, two sulfur-smelling compounds (**22** and **29**), a sweet-smelling compound (**53**), and an earthy-smelling compound (**56**) appeared with high FD factors. To identify the compounds responsible for the odors, first, the linear retention indices and odor quality of odor-active regions were compared to an in-house LRI and odor database; a chemical structure could be suggested for most of the odortants. Subsequently, the respective reference compounds were confirmed on the basis of odor quality and intensity matching at the sniffing port as well as mass spectra matching with the standard reference compounds. On the basis of this approach, 52 aroma active volatiles have been identified, including 17 ester compounds, 4 alcohol compounds, 5 acid compounds, 2 sulfur compounds, 2 aldehyde and ketone compounds, 5 furan compounds, 5 pyrazine compounds, 6 aromatic compounds, and 6 other compounds. The most odor-active compound was characterized as ethyl hexanoate. Ethyl hexanoate was confirmed as the key odorant for the overall aroma of roasted sesame-like flavor type liquor.[8] With somewhat lower FD factors, ethyl butanoate (**7**), ethyl 2-methylbutanoate (**9**), ethyl pentanoate (**12**), and ethyl 4-methylpentanoate (**14**) were suggested as potential contributors to the overall aroma.

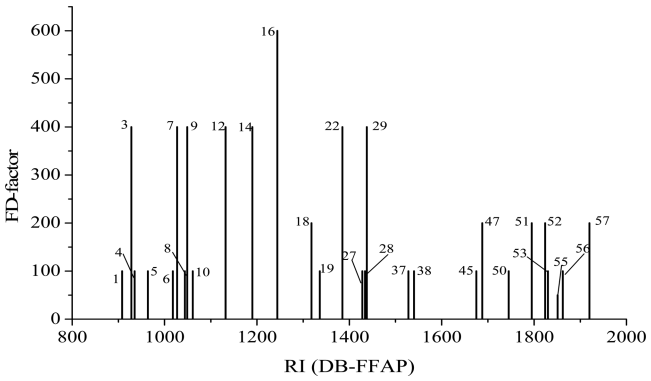


Figure 1. Flavor dilution chromatogram obtained by application of aroma extract dilution analysis (AEDA) on SP roasted sesame-like flavor. Odorants with an FD factor ≥50 are displayed. Numbering is identical with that in Table 1.

However, the concentrations of the remaining six sulfur odor-active regions were quite low and could be detected only with PFPD. These regions did not produce a MS TIC peak. To identify these six sulfur odor-active regions, first, the retention indices of odor-active area were compared to an in-house LRI and odor database; potential chemical structures for these odor regions could be suggested. Subsequently, multiple standards for each candidate sulfur volatile were run to find a retention time match by GC-PFPD. Finally, SIM MS confirmation with at least two characteristic ions was carried out by matching the LRI value and SIM MS spectrum with standard reference compounds. By this method, six sulfur aroma compounds were identified as dimethyl sulfide, *S*-methyl thioacetate, 2-furfurylthiol, ethyl 2-mercaptoacetate, methional, and ethyl 3-(methylthio)propanoate (Figure 3).

2-Furfurylthiol, exhibiting roasted sesame seed and coffee-like aromas, was one of the odor-active compounds with the highest FD (400) in SP liquor. 2-Furfurylthiol is a well-known powerful odorant that contributes to the characteristic aromas of various foods,[33−35] due to its low odor threshold (0.0025

Table 2. Calibration Curve Data of Sulfur-Containing Odorants and Their Recovery in Chinese Liquor

compound	slope	intercept	n^a	R^2	LOD^b (μg/L)	recovery (%)	linear range (μg/L)
dimethyl sulfide	0.45	0.62	7	0.9985	47	115	78−20000
S-methyl thioacetate	0.12	0.08	6	0.9919	3.1	95	74−4800
dimethyl trisulfide	0.05	−0.82	9	0.9901	0.27	78	59−30000
2-furfurylthiol	0.09	−0.01	8	0.9931	6.0	114	62−2000
methional	3.57	0.26	9	0.9921	2.7	91	80−10000
ethyl 2-mercaptoacetate	0.02	−0.01	7	0.9978	0.07	104	0.37−48
ethyl 3-(methylthio)propanoate	0.02	−0.01	8	0.9954	0.03	123	1.9−490
methionol	0.44	0.01	7	0.9995	0.45	106	14−3700

^{a}*n*, number of concentration gradients selected in standard curves. ^{b}LOD, limits of detection.

Downloaded from pubs.acs.org/jafcau/article-pdf/65/1/123/1411259/jf6b04242.pdf by JIANGNAN UNIV user on 18 October 2026

DOI: 10.1021/acs.jafc.6b04242
J. Agric. Food Chem. 2017, 65, 123−131

Table 3. Quantitative Data, Odor Thresholds, and OAVs of Odor-Active Compounds in SP Roasted Sesame-like Flavor Type Liquor

LRI	compound	odor threshold (μg/L)	average content (μg/L)	OAV[a]
1235	ethyl hexanoate	55[c]	627000 ± 2000[g]	11329
1409	ethyl octanoate	13[c]	98100 ± 200[g]	7608
1031	ethyl butanoate	82[c]	578000 ± 1000[g]	7094
915	3-methylbutanal	180[d]	22800 ± 200[i]	1339
1439	2-furfurylthiol	0.10[e]	118 ± 10[j]	1182
1128	ethyl pentanoate	27[c]	26900 ± 30[i]	1004
1060	ethyl 3-methylbutanoate	6.9[c]	6490 ± 400[i]	942
961	ethyl 2-methylpropanoate	58[c]	34500 ± 100[i]	600
1102	3-methylbutyl acetate	94[b]	46500 ± 800[i]	495
1555	2-methylpropanoic acid	1600[c]	357000 ± 1000[h]	226
1360	dimethyl trisulfide	0.36[d]	79 ± 5[j]	220
1045	ethyl 2-methylbutanoate	18[b]	2150 ± 100[i]	120
1837	β-damascenone	0.12[c]	14.0 ± 0.2[i]	116
1460	methional	7.1[b]	703 ± 20[j]	99
1602	butanoic acid	960[c]	78300 ± 300[h]	81
892	ethyl acetate	33000[c]	1830000 ± 50000[g]	56
1655	3-methylbutanoic acid	1100[c]	47400 ± 2000[h]	45
1727	pentanoic acid	390[c]	11600 ± 20[h]	30
1862	geosmin	0.11[f]	2.43 ± 0.02[i]	22
929	dimethyl sulfide	17[e]	237 ± 32[j]	14
1007	*S*-methyl thioacetate	21[e]	266 ± 6[j]	13
1768	ethyl 2-phenylacetate	410[c]	5050 ± 100[i]	12
1330	ethyl lactate	130000[c]	1460000 ± 100[g]	11
1080	2-methylpropanol	28000[c]	235000 ± 7000[g]	8
1039	1-propanol	54000[d]	350000 ± 9000[g]	6
1873	ethyl 3-phenylpropanoate	130[c]	661 ± 4[i]	5
1239	3-methylbutyl butanoate	920[d]	4020 ± 200[i]	4
1748	naphthalene	160[b]	557 ± 21[i]	3
1341	1-hexanol	5400[c]	13900 ± 200[g]	3
1846	hexanoic acid	2500[c]	6490 ± 600[h]	3
1456	furfural	44000[c]	110000 ± 2000[i]	2
1801	2-phenylethyl acetate	910[c]	2070 ± 50[i]	2
1454	3-methylbutyl hexanoate	1400[c]	2700 ± 20[i]	2
1424	acetic acid	160000[c]	280000 ± 1000[h]	2
2080	4-methylphenol	170[c]	266 ± 20[i]	2
953	ethyl propanoate	19000[c]	27500 ± 1000[i]	1
1642	ethyl decanoate	1100[c]	1090 ± 10[i]	1
2170	2-phenylethyl hexanoate	94[d]	79.4 ± 2[i]	1
1828	ethyl dodecanoate	640[b]	524 ± 20[i]	1
1958	2-phenylethyl butanoate	960[d]	527 ± 2[i]	1
1509	ethyl nonanoate	3200[c]	1630 ± 60[i]	1
1583	hexyl hexanoate	1900[d]	708 ± 64[i]	<1
1319	propyl hexanoate	13000[c]	2520 ± 63[i]	<1
1505	benzaldehyde	4200[c]	424 ± 9[i]	<1
1201	3-methylbutanol	180000[c]	126000 ± 3000[g]	<1
1336	ethyl heptanoate	13000[c]	9820 ± 400[g]	<1
1906	2-phenylethanol	29000[c]	3030 ± 200[i]	<1
1712	methionol	2100[b]	1880 ± 30[j]	<1
1356	2-methylpropyl hexanoate	5350.3[b]	1630 ± 9[i]	<1
1289	ethyl 2-furoate	130000[b]	659 ± 20[i]	<1
1380	2-nonanone	480[d]	233 ± 40[i]	<1
1640	ethyl benzoate	1400[c]	144 ± 20[i]	<1
1464	pentyl hexanoate	14000[c]	118 ± 6[i]	<1
2185	4-ethylphenol	620[d]	78.8 ± 10[i]	<1
1330	2,6-dimethylpyrazine	790[d]	536 ± 10[i]	<1
1425	2,3,5-trimethylpyrazine	730[d]	40.7 ± 0.6[i]	<1
1680	terpineol	2000[c]	6.18 ± 0.06[i]	<1
1557	ethyl 3-(methylthio)propanoate	3100[c]	56.9 ± 9[j]	<1
1670	3-methylbutyl octanoate	600[d]	54.1 ± 0.1[i]	<1
1857	furfuryl hexanoate	24000[b]	29.5 ± 0.2[i]	<1

Downloaded from pubs.acs.org/jafcau/article-pdf/65/1/123/1411259/jf6b04242.pdf by JIANGNAN UNIV user on 18 October 2026

DOI: 10.1021/acs.jafc.6b04242
J. Agric. Food Chem. 2017, 65, 123−131

Table 3. continued

LRI	compound	odor threshold (μg/L)	average content (μg/L)	OAV[a]
1602	2-undecanone	400[d]	11.1 ± 0.2[i]	<1
1368	nonanal	120[c]	10.8 ± 0.3[i]	<1
1518	ethyl 2-mercaptoacetate	120[e]	10.5 ± 0.6[j]	<1

[a]OAVs were calculated by dividing the concentration by odor threshold value of the compounds. [b]Odor threshold previously reported in ref 21. [c]Odor threshold previously reported in ref 20. [d]Odor threshold previously reported in ref 17. [e]Odor thresholds in 46% volume hydroalcoholic solution detected in this study. [f]Odor threshold previously reported in ref 43. [g]Quantified by GC-FID. [h]Quantified by LLM-GC-MS. [i]Quantified by HS-SPME-GC-MS. [j]Quantified by HS-SPME-GC-PFPD.

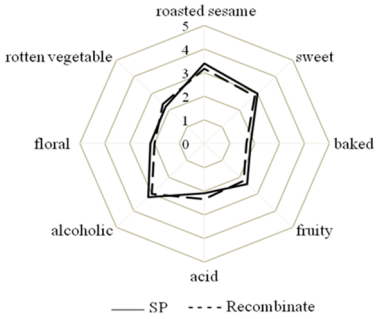


Figure 2. Aroma profiles of SP roasted sesame-like flavor type liquor and the aroma-reconstituted models containing those compounds with OAVs >1.0.

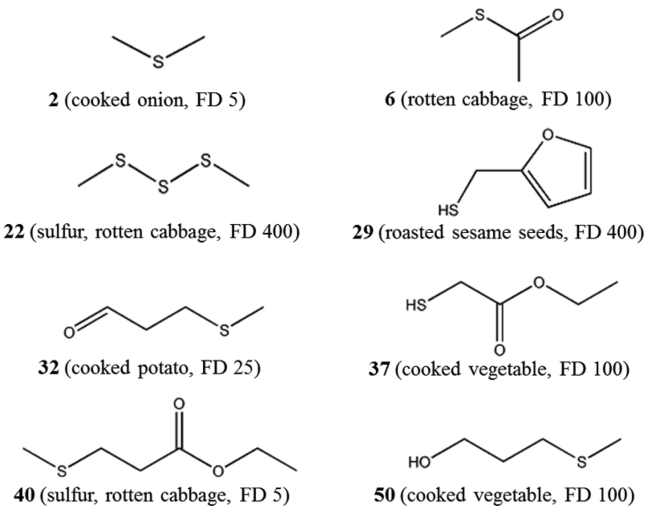


Figure 3. Structures of odor-active sulfur compounds identified in SP roasted sesame-like flavor type liquor (numbering refers to Table 1).

ng/L in air, 0.006 μg/L in water, 0.37 μg/kg in oil).[36,37] It was elucidated as a key odorant in roasted sesame seeds.[38,39] Because of its higher FD factor, 2-furfurylthiol might be a potent odor compound in roasted sesame-like flavor type liquors. Dimethyl trisulfide also had a high FD (400), showing sulfur and rotten cabbage aromas; it is an important odorant for the overall aroma of roasted sesame-like, strong, and soy sauce flavor type liquors.[8,17] With a somewhat lower FD factor (100), *S*-methyl thioacetate (rotten cabbage), ethyl 2-mercaptoacetate (cooked vegetable), and methionol (cooked vegetable) were suggested as potential contributors to the overall aroma of SP liquor. Methional exhibiting a cooked potato aroma also had a flavor dilution of 25, which was confirmed as a key odorant for the overall aroma of roasted sesame-like flavor type liquor.[8] In addition to these sulfur compounds, dimethyl sulfide and ethyl 3-(methylthio)propanoate were detected with low FD factors in SP liquor.

Two terpenoids were detected in the roasted sesame-like flavor type liquor with high FD factors for the first time, including β-damascenone (100) and geosmin (100). β-Damascenone, presenting honey and floral aromas, was previously reported as an important odorant in whiskey, rum, and brandy.[16,29,40] β-Damascenone was also identified as a key aroma in light and soy sauce flavor type Chinese liquors.[20,41]

Quantitation of Odor-Active Compounds and OAV Analysis. AEDA is a useful tool for screening aroma active compounds from the bulk of odorless volatiles; quantitative and OAV data are required for confirm the contribution of the odorants to roasted sesame-like flavor type liquor. Due to the complex chemical character and wide concentration range of the aroma active compounds identified in roasted sesame-like flavor type liquor, multiple quantitation approaches were employed in this study, including HS-SPME-GC-MS, LLME-GC-MS, GC-FID, and HS-SPME-GC-PFPD. The quantitation method for sulfur compounds was developed on the basis of HS-SPME-GC-PFPD (Table 2). A total of 65 compounds were quantitated in SP liquor. The highest concentration was determined for ethyl acetate, followed by ethyl lactate, ethyl hexanoate, and ethyl butanoate (Table 3). Other authors have reported lower concentrations of these compounds in roasted sesame-like flavor type liquors, which might be caused by the different liquor-making processes used.[8] For these sulfur compounds, methionol (1.8 mg/L) appeared with the highest concentration as expected. Ethyl 3-(methylthio)propanoate and ethyl 2-mercaptoacetate had lower concentrations of 10−60 μg/L.

To evaluate the contribution of quantified odorants to the overall aroma of the liquor, odor activity values (OAV, ratio of concentration to its odor threshold) were calculated for all compounds (Table 3). The calculation of OAVs suggested that 36 odorants should contribute to the overall aroma profile of SP liquor, because their concentrations exceeded their odor thresholds. A total of 21 esters were found with concentrations higher than their thresholds in SP liquor. Among them, ethyl hexanoate (OAV 11329), ethyl octanoate (OAV 7608), ethyl butanoate (OAV 7094), and ethyl pentanoate (OAV 1004) had the highest OAVs. These four esters were also previously identified as key odorants in roasted sesame-like flavor type liquor.[8] These results suggested the important aroma contribution of esters for SP liquor. Esters were also one of the most important aroma groups for other types of Chinese liquor, such as strong, light, soy sauce, and chixiang flavor type liquors.[17,20,21] Two of terpenoids identified in this study, β-damascenone and geosmin, were found with relatively high OAVs of 116 and 22, respectively, suggesting they had a certain aroma contribution to the roasted sesame-like flavor type liquor.

For the eight odor-active sulfur compounds identified in this study, five were quantified with OAVs >1. Due to the extremely

Downloaded from pubs.acs.org/jafcau/article-pdf/65/1/123/1411259/jf6b04242.pdf by JIANGNAN UNIV user on 18 October 2026

DOI: 10.1021/acs.jafc.6b04242
J. Agric. Food Chem. 2017, 65, 123−131

low odor threshold (0.1 μg/L) determined in this study, 2-furfurylthiol showed a very high OAV (1182) in the SP liquor. This indicated that 2-furfurylthiol had a strong contribution to the roasted sesame-like flavor type liquor. 2-Furfurylthiol, exhibiting intense roasted, coffee-like notes, was one of the key odorants for roasted sesame seeds.[38,39,42] Because roasted sesame-like aroma is the most important characteristic for roasted sesame-like flavor type liquor, 2-furfurylthiol might be a character-impact odorant for roasted sesame-like flavor type liquor. Dimethyl trisulfide (OAV 220) and methional (99) also showed important aroma contribution for SP liquor. This result agreed with Zheng et al.'s study previously carried out for roasted sesame-like flavor type liquor.[8] However, methionol, which was present in quite high concentration in SP liquor, showed OAV <1 and, thus, should not contribute to the SP liquor aroma as previously reported.[6,10]

Aroma Recombination Experiments. To validate the quantitative results, an aroma recombinate containing all key odorants at their actual concentrations measured in SP liquor (Table 3) was prepared in an ethanolic solution (53%, ethanol by volume). Descriptive profile tests were performed for the aroma recombinate in comparison to the SP liquor by rating the intensity of eight odor attributes as well as the overall similarity. The comparison of the aroma profiles of the aroma recombinate and SP liquor showed good similarity (Figure 2). The overall aroma similarities of the recombination model with the original liquor were judged to be 4.7 of 5.0 points. The aroma of the recombinate showed good similarity for roasted sesame, sweet, baked, fruity, acid, floral, alcoholic, and rotten vegetable. Consequently, this result indicated the success in simulating the typical aroma of the investigated liquor. To gain a deeper insight into the aroma contribution of 2-furfurylthiol, an omission experiment was carried out by a triangle test. Results showed that 9 of 10 assessors were able to detect the omission model lacking the 2-furfurythiol, thus leading to a very highly significant difference compared to the complete recombinate ($\alpha \leq 0.001$), thereby pinpointing the crucial role of this roasted sesame-smelling compound for the typical aroma of roasted sesame-like flavor type liquor. Consequently, the roasted sesame note present in the profile of overall aroma could be greatly linked to 2-furfurylthiol.

In summary, the aroma-active compounds in roasted sesame-like flavor type liquor were identified through HS-SPME-AEDA analysis, and 36 odorants were further screened as the important odorants according to quantitative study and OAV data. The aroma profile of roasted sesame-like flavor type liquor could be successfully reconstituted by mixing 36 aroma compounds in concentrations measured in this study. An omission experiment further confirmed 2-furfurylthiol could be a key odorant for the roasted sesame-like flavor type liquor.

AUTHOR INFORMATION

Corresponding Author

*(Y.X.) State Key Laboratory of Food Science & Technology, Key Laboratory of Industrial Biotechnology of Ministry of Education & School of Biotechnology, Jiangnan University, 1800 Lihu Avenue, Wuxi, Jiangsu, China 214122. Phone: +86-510-85964112. Fax: +86-510-85918201. E-mail: yxu@jiangnan.edu.cn.

ORCID

Yan Xu: 0000-0002-7919-4762

Funding

We gratefully acknowledge the National Natural Science Foundation of China (No. 31530055, 21506074), the Natural Science Foundation of Jiangsu Province (No. BK20140153), the National Key R&D Program (No. 2016YFD0400503), and the 111 Program of Introducing Talents (No. 111-2-06) for their financial support.

Notes

The authors declare no competing financial interest.

REFERENCES

(1) Li, P.; Lin, W. F.; Liu, X. Environmental factors affecting microbiota dynamics during traditional solid-state fermentation of Chinese daqu starter. *Front. Microbiol.* **2016**, *7*, 1237.

(2) Li, P.; Liang, H. B.; Lin, W. T. Microbiota dynamics associated with environmental conditions and potential roles of cellulolytic communities in traditional Chinese cereal starter solid-state fermentation. *Appl. Environ. Microbiol.* **2015**, *81*, 5144−5156.

(3) Fan, W. L.; Qian, M. C. Characterization of aroma compounds of Chinese "Wuliangye" and "Jiannanchun" liquors by aroma extract dilution analysis. *J. Agric. Food Chem.* **2006**, *54*, 2695−2704.

(4) Shen, Y. *Manual of Chinese Liquor Manufactures Technology*; Light Industry Publishing House of China: Beijing, China, 1996.

(5) Fan, W. L.; Xu, Y.; Qian, M. C. Identification of aroma compounds in Chinese "Moutai" and "Langjiu" liquors by normal phase liquid chromatography fractionation followed by gas chromatography/olfactometry. In *Flavor Chemistry of Wine and Other Alcoholic Beverages*; Qian, M. C., Shellhammer, T. H., Eds.; American Chemical Society: Washington, DC, USA, 2012; pp303−338.

(6) Wang, H. P.; Li, Z. P. Research seminar of "sesame" flavor liquor. *Liquor Making Sci. Technol.* **1986**, *2*, 17−18.

(7) Zhou, Q. Y.; Fan, W. L.; Xu, Y. Important volatile aroma compounds in Chinese roasted-sesame-like aroma type Jingzhi liquors. *Sci. Technol. Food Ind.* **2015**, *36*, 62−67.

(8) Zheng, Y.; Sun, B. G.; Zhao, M. M.; Zheng, F. P.; Huang, M. Q.; Sun, J. Y.; Sun, X. T.; Li, H. H. Characterization of the key odorants in Chinese zhima aroma-type baijiu by gas chromatography-olfactometry, quantitative measurements, aroma recombination, and omission studies. *J. Agric. Food Chem.* **2016**, *64*, 5367−5374.

(9) Zhang, Y. Y.; Sun, J. Y.; Zhang, F. G. Determination of sulfur compunds in bandaojing sesame-flavor liquor. *J. Chinese Inst. Food Sci. Technol.* **2012**, *12*, 173−179.

(10) Zhang, Y. Y.; Sun, J. Y.; Zhang, F. G. Analysis of sulfur compounds in sesame flavor liquor. *J. Chinese Inst. Food Sci. Technol.* **2014**, *5*, 218−225.

(11) Dunkel, A.; Steinhaus, M.; Kotthoff, M.; Nowak, B.; Krautwurst, D.; Schieberle, P.; Hofmann, T. Nature's chemical signatures in human olfaction: a foodborne perspective for future biotechnology. *Angew. Chem., Int. Ed.* **2014**, *53*, 7124−7143.

(12) Grosch, W. Determination of potent odourants in foods by aroma extract dilution analysis (AEDA) and calculation of odour activity values (OAVs). *Flavour Fragrance J.* **1994**, *9*, 147−158.

(13) Christlbauer, M.; Schieberle, P. Characterization of the key aroma compounds in beef and pork vegetable gravies a la chef by application of the aroma extract dilution analysis. *J. Agric. Food Chem.* **2009**, *57*, 9114−9122.

(14) Schnermann, P.; Schieberle, P. Evaluation of key odorants in milk chocolate and cocoa mass by aroma extract dilution analyses. *J. Agric. Food Chem.* **1997**, *45*, 867−872.

(15) Franitza, L.; Granvogl, M.; Schieberle, P. Characterization of the key aroma compounds in two commercial rums by means of the sensomics approach. *J. Agric. Food Chem.* **2016**, *64*, 637−645.

(16) Willner, B.; Granvogl, M.; Schieberle, P. Characterization of the key aroma compounds in Bartlett pear brandies by means of the sensomics concept. *J. Agric. Food Chem.* **2013**, *61*, 9583−9593.

(17) Wang, X.; Fan, W.; Xu, Y. Comparison on aroma compounds in Chinese soy sauce and strong aroma type liquors by gas

Downloaded from pubs.acs.org/jafcau/article-pdf/65/1/123/1411259/jf6b04242.pdf by JIANGNAN UNIV user on 18 October 2026

DOI: 10.1021/acs.jafc.6b04242
J. Agric. Food Chem. 2017, 65, 123−131

chromatography-olfactometry, chemical quantitative and odor activity values analysis. *Eur. Food Res. Technol.* **2014**, *239*, 813−825.
(18) Fan, W. L.; Qian, M. C. Headspace solid phase microextraction and gas chromatography-olfactometry dilution analysis of young and aged Chinese "Yanghe Daqu" liquors. *J. Agric. Food Chem.* **2005**, *53*, 7931−7938.
(19) Du, X. F.; Song, M.; Rouseff, R. Identification of new strawberry sulfur volatiles and changes during maturation. *J. Agric. Food Chem.* **2011**, *59*, 1293−1300.
(20) Gao, W. J.; Fan, W. L.; Xu, Y. Characterization of the key odorants in light aroma type Chinese liquor by gas chromatography-olfactometry, quantitative measurements, aroma recombination, and omission studies. *J. Agric. Food Chem.* **2014**, *62*, 5796−5804.
(21) Fan, H. Y.; Fan, W. L.; Xu, Y. Characterization of key odorants in Chinese Chixiang aroma-type liquor by gas chromatography-olfactometry, quantitative measurements, aroma recombination, and omission studies. *J. Agric. Food Chem.* **2015**, *63*, 3660−3668.
(22) Feng, Y. Z.; Cai, Y.; Su, G. W.; Zhao, H. F.; Wang, C. X.; Zhao, M. M. Evaluation of aroma differences between high-salt liquid-state fermentation and low-salt solid-state fermentation soy sauces from China. *Food Chem.* **2014**, *145*, 126−134.
(23) Cates, V. E.; Meloan, C. E. Separation of sulfones by gas chromatography. *J. Chromatogr. A* **1963**, *11*, 472−480.
(24) AQSIQ. *SAC Method of Analysis for Chinese Spirits*; Standardization Publishing House of China: Beijing, China, 2007; GB/T 10345-2007, pp 1−17.
(25) Vazquez-Landaverde, P. A.; Torres, J. A.; Qian, M. C. Effect of high-pressure-moderate-temperature processing on the volatile profile of milk. *J. Agric. Food Chem.* **2006**, *54*, 9184−9192.
(26) Fang, Y.; Qian, M. C. Sensitive quantification of sulfur compounds in wine by headspace solid-phase microextraction technique. *J. Chromatogr. A* **2005**, *1080*, 177−185.
(27) Burbank, H.; Qian, M. C. Development of volatile sulfur compounds in heat-shocked and pasteurized milk cheese. *Int. Dairy J.* **2008**, *18*, 811−818.
(28) Czerny, M.; Christlbauer, M.; Christlbauer, M.; Fischer, A.; Granvogl, M.; Hammer, M.; Hartl, C.; Hernandez, N.; Schieberle, P. Re-investigation on odour thresholds of key food aroma compounds and development of an aroma language based on odour qualities of defined aqueous odorant solutions. *Eur. Food Res. Technol.* **2008**, *228*, 265−273.
(29) Poisson, L.; Schieberle, P. Characterization of the key aroma compounds in an American Bourbon whisky by quantitative measurements, aroma recombination, and omission studies. *J. Agric. Food Chem.* **2008**, *56*, 5820−5826.
(30) Jellinek, G. *Evaluation of Food. Theory and Practice*; Horwood: Chichester, UK, 1985.
(31) Acena, L.; Vera, L.; Guasch, J.; Busto, O.; Mestres, M. Determination of roasted pistachio (*Pistacia vera* L.) key odorants by headspace solid-phase microextraction and gas chromatography olfactometry. *J. Agric. Food Chem.* **2011**, *59*, 2518−2523.
(32) Baek, H. H.; Kim, H. J. Solid phase microextraction-gas chromatography-olfactometry of soy sauce based on sample dilution analysis. *Food Sci. Biotechnol.* **2004**, *13*, 90−95.
(33) Burdack-Freitag, A.; Schieberle, P. Characterization of the key odorants in raw Italian hazelnuts (*Corylus avellana* L. var. Tonda Romana) and roasted hazelnut paste by means of molecular sensory science. *J. Agric. Food Chem.* **2012**, *60*, 5057−5064.
(34) Grosch, W. Evaluation of the key odorants of foods by dilution experiments, aroma models and omission. *Chem. Senses* **2001**, *26*, 533−545.
(35) Hofmann, T.; Czerny, M.; Calligaris, S.; Schieberle, P. Model studies on the influence of coffee melanoidins on flavor volatiles of coffee beverages. *J. Agric. Food Chem.* **2001**, *49*, 2382−2386.
(36) Buttery, R. G.; Ling, L. C.; Stern, D. J. Studies on popcorn aroma and flavor volatiles. *J. Agric. Food Chem.* **1997**, *45*, 837−843.
(37) Rychlik, M.; Schieberle, P.; Grosch, W. *Compilation of Odor Thresholds, Odor Qualities and Retention Indices of Key Food Odorants*; Deutsche Forschungsanstalt für Lebensmittelchemie und Institut für Lebensmittelchemie der TU München: Garching, Germany, 1998.
(38) Schieberle, P. Odour-active compounds in moderately roasted sesame. *Food Chem.* **1996**, *55*, 145−152.
(39) Tamura, H.; Fujita, A.; Steinhaus, M.; Takahisa, E.; Watanabe, H.; Schieberle, P. Identification of novel aroma-active thiols in pan-roasted white sesame seeds. *J. Agric. Food Chem.* **2010**, *58*, 7368−7375.
(40) Pino, J. A.; Tolle, S.; Gok, R.; Winterhalter, P. Characterisation of odour-active compounds in aged rum. *Food Chem.* **2012**, *132*, 1436−1441.
(41) Wang, L.; Hu, G. Y.; Lei, L. B.; Lin, L.; Wang, D. Q.; Wu, J. X. Identification and aroma impact of volatile terpenes in Moutai liquor. *Int. J. Food Prop.* **2016**, *19*, 1335−1352.
(42) Grosch, W. Flavour of coffee. A review. *Nahrung* **1998**, *42*, 344−350.
(43) Du, H.; Fan, W. L.; Xu, Y. Characterization of geosmin as source of earthy odor in different aroma type Chinese liquors. *J. Agric. Food Chem.* **2011**, *59*, 8331−8337.

Downloaded from pubs.acs.org/jafcau/article-pdf/65/1/123/1411259/jf6b04242.pdf by JIANGNAN UNIV user on 18 October 2026

DOI: 10.1021/acs.jafc.6b04242
J. Agric. Food Chem. 2017, 65, 123−131